

DEPARTMENT OF MATHEMATICS, IIT - GUWAHATI

Odd Semester of the Academic Year 2023-2024

MA 101 Mathematics II

Problem Sheet 3: Critical points, maxima and minima, Lagrange's multipliers and double integrals.

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1. Find the local maximum and minimum values and saddle point(s) of the functions:

(a) $f(x, y) = x^2 + y^2 + x^2y + 4$

(b) $f(x, y) = 4xy - x^4 - y^4$

(c) $f(x, y) = \sin x \cosh y$

(d) $f(x, y) = x + 2y + \frac{4}{x} - y^2$.

Solution: (a) Since $f(x, y) = x^2 + y^2 + x^2y + 4$,
 $f_x(x, y) = 2x + 2xy$ and $f_y(x, y) = 2y + x^2$.

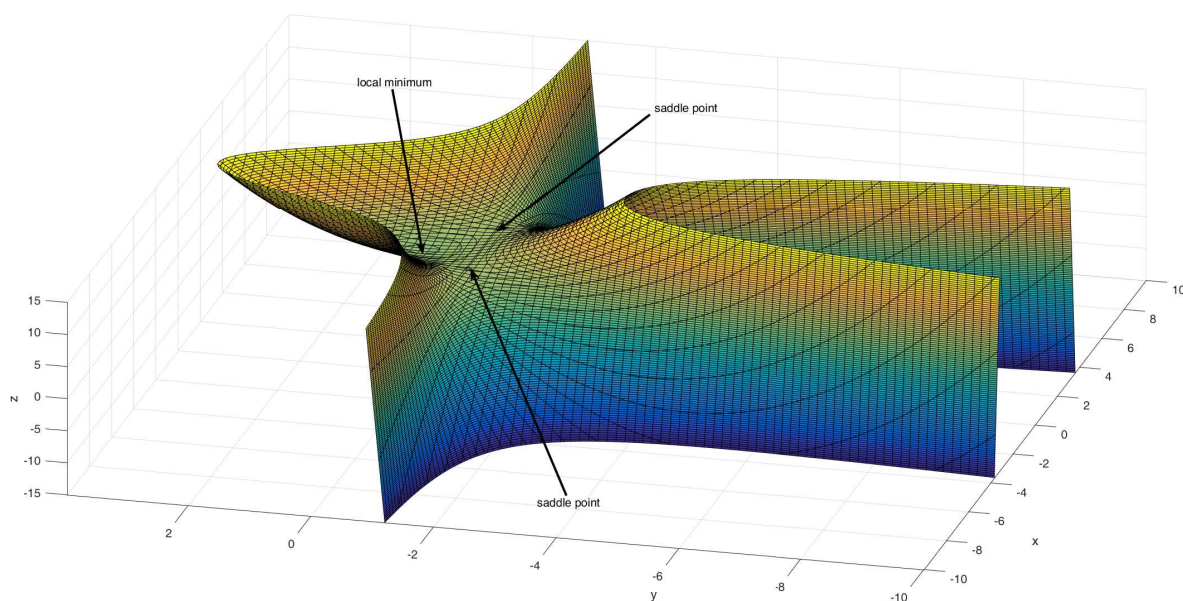
The critical points are given by the following equations:

$f_x(x, y) = f_y(x, y) = 0$, which gives $x(y + 1) = 0$ and $2y = -x^2$,
 $\Rightarrow (0, 0)$ and $(\sqrt{2}, -1), (-\sqrt{2}, -1)$ are the three critical points.

$D(x, y) = f_{xx}f_{yy} - f_{xy}^2 = (2 + 2y)(2) - (2x)^2 = 4(1 + y - x^2)$.

At $(0, 0)$, $D(0, 0) = 4 > 0$, and $f_{xx}(0, 0) = 2 > 0$ hence $(0, 0)$ is a point of local minimum.

Since $D(\sqrt{2}, -1) = D(-\sqrt{2}, -1) < 0$, $(\sqrt{2}, -1), (-\sqrt{2}, -1)$ are saddle points of f .



(b) Since $f(x, y) = 4xy - x^4 - y^4$,
 $f_x(x, y) = 4y - 4x^3$ and $f_y(x, y) = 4x - 4y^3$.

The critical points are given by the following equations:

$$f_x(x, y) = f_y(x, y) = 0, \text{ which gives } y = x^3 \text{ and } x = y^3, \text{ or } x = x^9, \\ \Rightarrow x(x^4 - 1)(x^4 + 1) = 0. \quad (1)$$

The only real roots to (1) are $x = 0, +1, -1$,

$\Rightarrow$ the critical points are $(0, 0), (1, 1), (-1, -1)$.

$$D(x, y) = f_{xx}f_{yy} - f_{xy}^2 = (-12x^2)(-12y^2) - 4^2 = 144x^2y^2 - 16.$$

At $(0, 0)$, $D(0, 0) = -16 < 0$, hence $(0, 0)$ is a saddle point.

At $(1, 1)$ and $(-1, -1)$, $D(1, 1) = D(-1, -1) > 0$, and $f_{xx}(1, 1) = f_{xx}(-1, -1) = -12$ hence $(1, 1)$ and $(-1, -1)$ are both points of local maxima.

(c) $f_x(x, y) = \cos x \cosh y$ and $f_y(x, y) = \sin x \sinh y$.

The critical points are given by the following equations:

$$f_x(x, y) = f_y(x, y) = 0, \text{ which gives } x = \frac{(2n+1)\pi}{2}, y = 0 \text{ for } n = 0, \pm 1, \pm 2, \dots$$

$$f_{xx}(x, y) = -\sin x \cosh y, f_{yx}(x, y) = \cos x \sinh y, f_{yy}(x, y) = \sin x \cosh y.$$

$$D\left(\frac{(2n+1)\pi}{2}, 0\right) = f_{xx}(0, 0)f_{yy}(0, 0) - (f_{xy}(0, 0))^2 < 0, \text{ hence } \left(\frac{(2n+1)\pi}{2}, 0\right) \text{ for } \\ n = 0, \pm 1, \pm 2 \text{ are all saddle points of } f.$$

(d) Since $f(x, y) = x + 2y + \frac{4}{x} - y^2$,

$$f_x(x, y) = 1 - 4x^{-2} \text{ and } f_y(x, y) = 2 - 2y, \text{ (for } x \neq 0\text{)}.$$

The critical points are given by the following equations:

$$f_x(x, y) = f_y(x, y) = 0, \text{ which gives } x = \pm 2, y = 1.$$

$\Rightarrow$ the critical points are $(2, 1), (-2, 1)$.

$$D(x, y) = f_{xx}f_{yy} - f_{xy}^2 = \left(\frac{8}{x^3}\right)(-2) - 0,$$

$$\Rightarrow D(2, 1) < 0 \text{ and } D(-2, 1) > 0, \text{ and } f_{xx}(-2, 1) < 0.$$

Hence $(2, 1)$ is a saddle point and $(-2, 1)$ is a point of local maximum.

- Find the absolute maximum and minimum values of $f(x, y) = 4xy^2 - x^2y^2 - xy^3$ on the set D where D is the closed triangular region in the xy -plane with vertices $(0, 0)$, $(0, 6)$ and $(6, 0)$.

Solution: Since $f(x, y) = 4xy^2 - x^2y^2 - xy^3$, $f_x(x, y) = 4y^2 - 2xy^2 - y^3$, $f_y(x, y) = 8xy - 2x^2y - 3xy^2$.

The critical points are given by the following equations:

$$f_x(x, y) = f_y(x, y) = 0, \text{ which gives } y^2(4 - 2x - y) = 0 \text{ and } xy(8 - 2x - 3y) = 0.$$

Hence the critical points are given by the conditions:

(i) $y = 0, x = 0$, $(0, 0)$ on the boundary of D

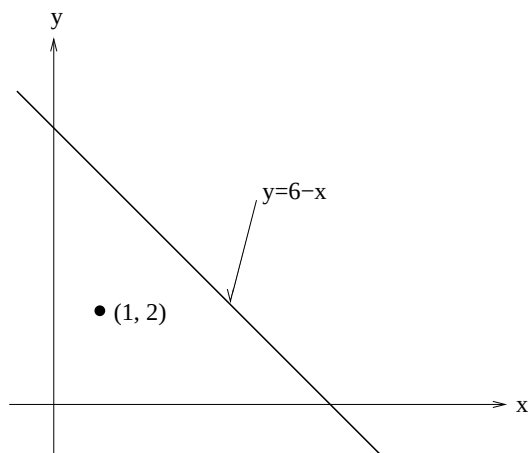
(ii) $y = 0, 8 - 2x = 0$, which gives $(4, 0)$ on the boundary of D

(iii) $4 - 2x - y = 0, x = 0$, which gives the point $(0, 4)$ on the boundary of D

(iv) $4 - 2x - y = 0, 8 - 2x - 3y = 0$ which gives the point $(1, 2)$, which is in the interior of the triangle D

(v) $y = 0, 4 - 2x = 0$ which gives $(2, 0)$ which is on the boundary of D .

$$f_{xx}(x, y) = -2y^2, f_{xy}(x, y) = 8y - 4xy - 3y^2, \text{ and } f_{yy}(x, y) = 8x - 2x^2 - 6xy.$$



At $(1, 2)$, $D(1, 2) = (-8)(8 - 2 - 12) - (16 - 8 - 12)^2 = 48 - 16 = 32 > 0$ and $f_{xx}(1, 2) = -8$, hence $(1, 2)$ is a point of local maxima and $f(1, 2) = 4$.

On the boundary $y = 0$ of D , $f(x, y) = 0$.

On the boundary $x = 0$, $f(x, y) = 0$.

On the boundary $y = 6 - x$, $f(x, y) = 4x(6 - x)^2 - x^2(6 - x)^2 - x(6 - x)^3 = -2x(6 - x)^2$.

The critical points of f on the line $y = 6 - x$ is given by:

$$f_x(x, 6 - x) = 0 = -6(x - 2)(x - 6),$$

$\Rightarrow x = 2, y = 4$, and $x = 6, y = 0$ are the only critical points of f on this line.

$$f_{xx}(x, 6 - x) = -6(2x - 8), \text{ hence for } x = 2, f_{xx}(2, 4) = 24 > 0,$$

$\Rightarrow (2, 4)$ is a point of local minimum on the line.

For $x = 6$, $f(6, 0) = 0$.

The possible candidates for absolute maxima are $(1, 2)$ and points on the line $x = 0$ and $y = 0$, but $f(1, 2) = 4 > f(x, 0) = f(0, y) = 0$, hence $(1, 2)$ is a point of absolute maximum.

The possible candidates for absolute minima are $(2, 4)$ and points on the line $x = 0$ and $y = 0$, but $f(2, 4) = -64 < f(x, 0) = f(0, y) = 0$, hence $(2, 4)$ is a point of absolute minimum.

3. Check that for the following functions the origin is a critical point; determine whether $f(\mathbf{0})$ is a local minimum value, a local maximum value or neither.

(a) $f(x, y, z) = 5x^2 + 4y^2 + 7z^2 + 4xy + 2z \sin x + 6y \sin z$

(b) $f(w, x, y, z) = wx + 2xy + 3yz - w^2 - 2x^2 - 3y^2 - 4z^2$.

Solution:

(a) The conditions $f_x(x, y, z) = 10x + 4y + 2z \cos x = 0$, $f_y(x, y, z) = 8y + 4x + 6 \sin z = 0$, $f_z(x, y, z) = 14z + 2 \sin x + 6y \cos z = 0$, are satisfied by $(0, 0, 0)$, hence it is a critical point of f .

$$f_{xx}(x, y, z) = 10 - 2z \sin x, f_{yy}(x, y, z) = 8, f_{zz}(x, y, z) = 14 - 6y \sin z,$$

$$f_{yx}(x, y, z) = 4, f_{zy}(x, y, z) = 6 \cos z, f_{xz}(x, y, z) = 2 \cos x,$$

$$f_{xx}(0, 0, 0) = 10, f_{yy}(0, 0, 0) = 8, f_{zz}(0, 0, 0) = 14,$$

$$f_{yx}(0, 0, 0) = 4, f_{zy}(0, 0, 0) = 6, f_{xz}(0, 0, 0) = 2.$$

Hence the Hessian matrix is given by

$$H = \begin{bmatrix} 10 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 14 \end{bmatrix}.$$

Since $\Delta_1 = f_{xx}(0, 0, 0) = 10 > 0$, $\Delta_2 = f_{xx}(0, 0, 0)f_{yy}(0, 0, 0) - (f_{xy}(0, 0, 0))^2 = 64 > 0$ and $\Delta_3 = |H| = 600 > 0$, hence $(0, 0, 0)$ is a point of local minimum.

(b) $f(w, x, y, z) = wx + 2xy + 3yz - w^2 - 2x^2 - 3y^2 - 4z^2$

$$f_w(x, y, z, w) = x - 2w, f_{ww}(x, y, z, w) = -2, f_{xw}(x, y, z, w) = 1, f_{yw}(x, y, z, w) = 0, f_{zw}(x, y, z, w) = 0.$$

$$f_x(x, y, z, w) = w + 2y - 4x, f_{xx}(x, y, z, w) = -4, f_{yx}(x, y, z, w) = 2, f_{zx}(x, y, z, w) = 0$$

$$f_y(x, y, z, w) = 2x + 3z - 6y, f_{yy}(x, y, z, w) = -6, f_{zy}(x, y, z, w) = 3$$

$$f_z(x, y, z, w) = 3y - 8z, f_{zz}(x, y, z, w) = -8.$$

Since $f_w(x, y, z, w) = f_x(x, y, z, w) = f_y(x, y, z, w) = f_z(x, y, z, w) = 0$, at $(0, 0, 0, 0)$ hence it is a critical point of f and the Hessian matrix at $(0, 0, 0, 0)$ is given by

$$H = \begin{bmatrix} -2 & 1 & 0 & 0 \\ 1 & -4 & 2 & 0 \\ 0 & 2 & -6 & 3 \\ 0 & 0 & 3 & -8 \end{bmatrix}.$$

Therefore $\Delta_1 = -2 < 0$, $\Delta_2 = 7 > 0$, $\Delta_3 = -34 < 0$, $\Delta_4 = 337 > 0$.

Since $(-1)^k \Delta_k > 0$ for all $k = 1, 2, 3, 4$, $(0, 0, 0, 0)$ is a local maximum point of f .

4. Find the points on the surface $z^2 = xy + 1$ that are closest to the origin.

Solution: The square of the distance of any point of the given surface from the origin is given by:

$$f(x, y) = x^2 + y^2 + z^2 = x^2 + y^2 + xy + 1$$

To find the absolute minimum of f .

The critical points are given by the following equations:

$f_x(x, y) = f_y(x, y) = 0$, which gives $2x + y = 0 = 2y + x$, the only solution to this system is $(0, 0)$.

$D(0, 0) = f_{xx}(0, 0)f_{yy}(0, 0) - (f_{xy}(0, 0))^2 = 2 \times 2 - 1 = 3 > 0$ and $f_{xx}(0, 0) = 2 > 0$, hence $(0, 0)$ is a point of local minimum.

Since $(0, 0)$ is the only critical point so $(0, 0)$ is the point of absolute minimum of f .

5. Find the volume of the largest rectangular box with edges parallel to the axes that can be inscribed in the ellipsoid $9x^2 + 36y^2 + 4z^2 = 36$.

Solution: Since the ellipsoid is centered at $(0, 0, 0)$, clearly the rectangular box of maximum volume should also be centered at the origin and if (x, y, z) gives a corner of the rectangle in the first octant, which touches the ellipsoid then clearly the other corners are $(-x, y, z), (x, -y, z), (x, y, -z), (x, -y, -z), (-x, -y, z), (-x, y, -z)$ and $(-x, -y, -z)$. Then the volume of the rectangular box is given by

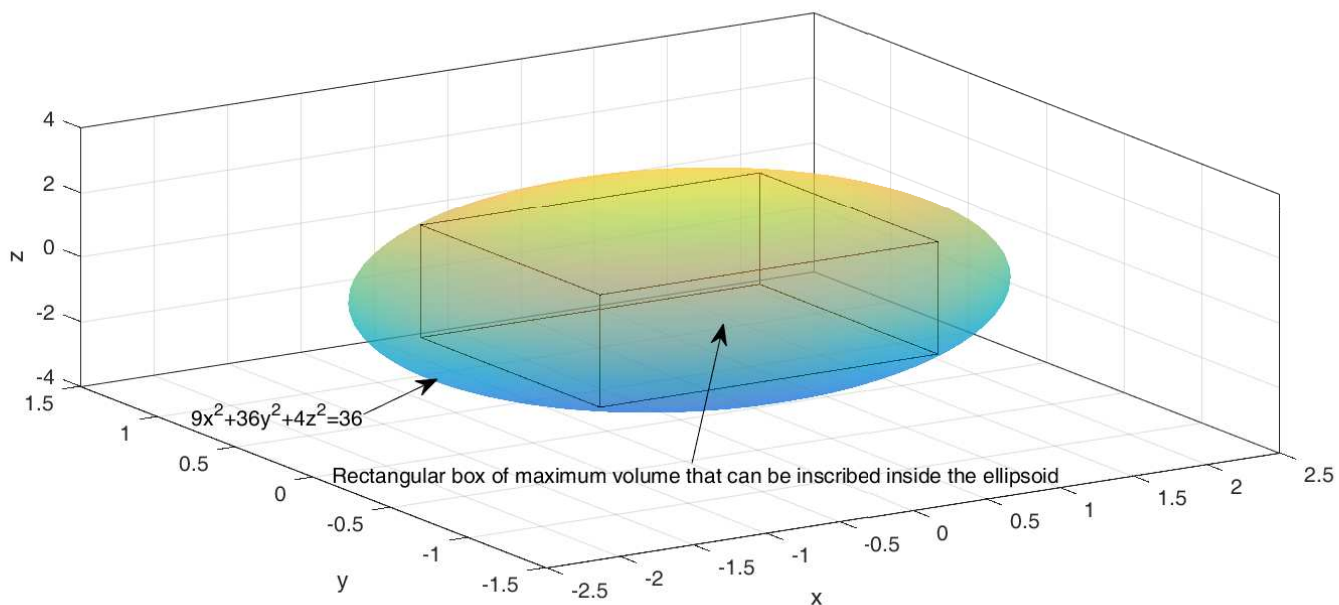
$V = 8xyz$ and the point (x, y, z) should satisfy the condition $9x^2 + 36y^2 + 4z^2 = 36$.

So the problem is to max $8xyz$ or max $cx^2y^2z^2$ ($c > 0$)

subject to $4z^2 = 36 - 9x^2 - 36y^2$.

or max $x^2y^2(36 - 9x^2 - 36y^2)$.

The critical points are given by the following conditions:



$$f_x(x, y) = (2x)y^2(36 - 9x^2 - 36y^2) + x^2y^2(-18x) = 0 \text{ or } xy^2(2 - x^2 - 2y^2) = 0 \text{ and}$$

$$f_y(x, y) = (2y)x^2(36 - 9x^2 - 36y^2) + x^2y^2(-72y) = 0 \text{ or } yx^2(4 - x^2 - 8y^2) = 0.$$

The critical points satisfy either of the following conditions:

(i) $x = 0$

(ii) $y = 0$

(iii) $x \neq 0, y \neq 0, 2 - x^2 - 2y^2 = 0$ and $4 - x^2 - 8y^2 = 0$, which gives $y = \pm \frac{1}{\sqrt{3}}$ and $x = \pm \frac{2}{\sqrt{3}}$.

Since for critical points of the type (i), (ii), $V = 0$,

we need to check for solutions of type (iii) if it maximizes V .

(**) Since $D(x, y) = f_{xx}(x, y)f_{yy}(x, y) - f_{xy}(x, y)^2 > 0$, and $f_{xx}(x, y) < 0$ at $(\pm \frac{2}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}})$ these points give the (x, y) coordinates of the corner points of the inscribed rectangular solid of maximum volume.

Since the z coordinates of the solid are $\pm\sqrt{3}$, and the volume of the required solid is $8 \times \frac{2}{\sqrt{3}} \times \frac{1}{\sqrt{3}} \times \sqrt{3}$.

(**) **Aliter:** Since the existence of such a solid is guaranteed (maximizing a continuous function $x^2y^2z^2$ in a closed bounded region) so the critical points other than those given by $x = 0, y = 0$ (which gives the minimum value of $x^2y^2z^2$) must necessarily maximize $x^2y^2z^2$.

Aliter: Using lagrange's multipliers to solve the above problem we get:

$$\max x^2y^2z^2$$

$$\text{subject to } g(x, y, z) = 4z^2 - 36 + 9x^2 + 36y^2 = 0.$$

Let λ such that

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z)$$

$$\Rightarrow (2xy^2z^2, 2yx^2z^2, 2zy^2x^2) = \lambda(18x, 72y, 8z) \quad (1)$$

$\Rightarrow$ either $x = 0, y = 0$ or $z = 0$.

or none of $x = 0, y = 0, z = 0$ is satisfied, then $\lambda \neq 0$ and (1) implies:

$$y^2 z^2 = 9\lambda, \quad x^2 z^2 = 36\lambda \quad \text{and} \quad y^2 x^2 = 4\lambda,$$

$$\Rightarrow x^4 = 16\lambda \quad \text{or} \quad x^2 = 4\sqrt{\lambda}, \quad \text{which implies,}$$

$$y^2 = \sqrt{\lambda}, \quad z^2 = 9\sqrt{\lambda}, \quad \text{which when substituted in } g(x, y, z) = 0 \text{ gives } \sqrt{\lambda} = \frac{1}{3}.$$

$$\text{Hence } x = \pm \frac{2}{\sqrt{3}}, \quad y = \pm \frac{1}{\sqrt{3}}, \quad z = \pm \sqrt{3}.$$

6. The plane $x + y + z = 12$ cuts the paraboloid $z = x^2 + y^2$ in an ellipse. Find the highest and lowest points on this ellipse.

Solution: The required problem is to Maximize or Minimize $f(x, y, z) = z$

subject to $g(x, y, z) = x + y + z - 12 = 0$ and $h(x, y, z) = x^2 + y^2 - z = 0$.

Using Lagrange's method, let λ and μ be such that

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z) + \mu \nabla h(x, y, z)$$

$$\Rightarrow (0, 0, 1) = \lambda(1, 1, 1) + \mu(2x, 2y, -1)$$

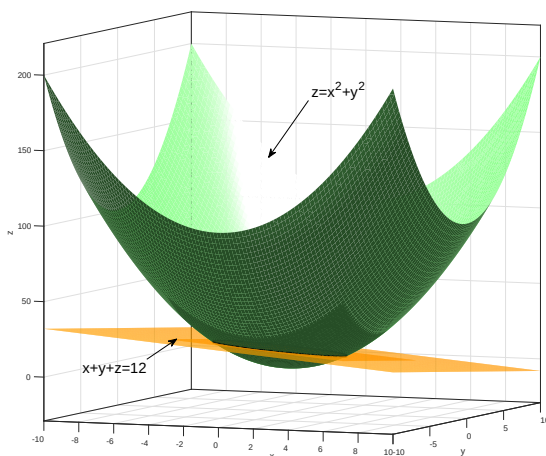
$$\Rightarrow \lambda + 2\mu x = 0, \quad \lambda + 2\mu y = 0 \quad \text{and} \quad \lambda - \mu = 1, \quad \text{which gives } 2\mu(x - y) = 0.$$

Since $\mu = 0$ implies $\lambda = 0$ which contradicts $\lambda - \mu = 1$, hence $x = y$.

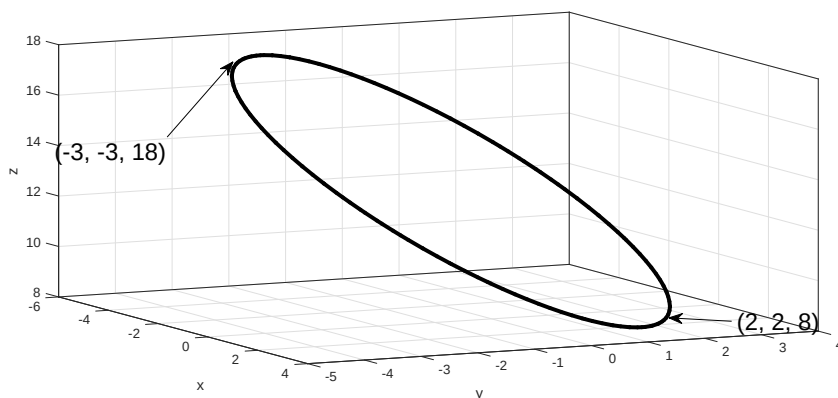
When we substitute $y = x$ in the equation for the plane and the paraboloid we get

$$12 - 2x = z = 2x^2, \quad \text{which gives } (x - 2)(x + 3) = 0,$$

so the critical points are $(2, 2, 8)$, the lowest point and $(-3, -3, 18)$, the highest point.



(a)



(b)

7. Use Lagrange multipliers to find the global minimum and maximum values of the functions subject to the given constraint(s)

(a) $f(x, y) = 4x + 6y; x^2 + y^2 = 13$

(b) $f(x_1, x_2, \dots, x_n) = x_1 + x_2 + \dots + x_n; x_1^2 + x_2^2 + \dots + x_n^2 = 1$

(c) $f(x, y) = e^{-xy}; x^2 + 4y^2 \leq 1.$

Solution:

(a) Maximize or Minimize $f(x, y) = 4x + 6y$ subject to $g(x, y) = x^2 + y^2 - 13 = 0$.
Let λ be such that

$$\nabla f(x, y) = \lambda \nabla g(x, y)$$

$$\Rightarrow (4, 6) = \lambda(2x, 2y) \Rightarrow \lambda x = 2, \lambda y = 3$$

$$\Rightarrow 2y = 3x, \text{ and } g(x, \frac{3}{2}x) = 13x^2 - 13 \times 4 = 0 \text{ implies } x = 2, y = 3,$$

$$\text{or } x = -2, y = -3.$$

Since $(2, 3)$ and $(-2, -3)$ are the only two solutions of the above equations:

$f(2, 3) = 26$ and $f(-2, -3) = -26$ gives the absolute minimum and maximum values of f .

(b) Maximize or Minimize $f(x_1, x_2, \dots, x_n) = x_1 + x_2 + \dots + x_n$ subject to $g(x_1, x_2, \dots, x_n) = x_1^2 + x_2^2 + \dots + x_n^2 - 1 = 0$.

Let λ be such that

$$\nabla f(x_1, x_2, \dots, x_n) = \lambda \nabla g(x_1, x_2, \dots, x_n).$$

$$(1, 1, \dots, 1) = \lambda(2x_1, 2x_2, \dots, 2x_n)$$

$$\Rightarrow x_1 = x_2 = \dots = x_n, \text{ and } g(x_1, x_1, \dots, x_1) = nx_1^2 - 1 = 0 \text{ gives}$$

$$x_i = \frac{1}{\sqrt{n}} \text{ for all } i = 1, 2, \dots, n \text{ or } x_i = -\frac{1}{\sqrt{n}} \text{ for all } i = 1, 2, \dots, n.$$

The absolute maximum is attained at $(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}, \dots, \frac{1}{\sqrt{n}})$ and the absolute minimum is attained at $(-\frac{1}{\sqrt{n}}, -\frac{1}{\sqrt{n}}, \dots, -\frac{1}{\sqrt{n}})$.

(c) Maximize or Minimize $f(x, y) = e^{-xy}$ subject to $x^2 + 4y^2 \leq 1$.

The conditions for critical points inside the ellipse $x^2 + 4y^2 = 1$ are given by:

$$f_x(x, y) = 0 \text{ and } f_y(x, y) = 0 \text{ which gives } -ye^{-xy} = 0 \text{ and } -xe^{-xy} = 0,$$

$$\Rightarrow x = 0 \text{ and } y = 0, \text{ which lies inside the ellipse.}$$

$D(0, 0) = f_{xx}(0, 0)f_{yy}(0, 0) - f_{xy}(0, 0)^2 = -e^0 = -1 < 0$, hence $(0, 0)$ is a saddle point of f .

For points on the boundary of the ellipse $x^2 + 4y^2 = 1$, we use Lagrange's method.

Let λ be such that

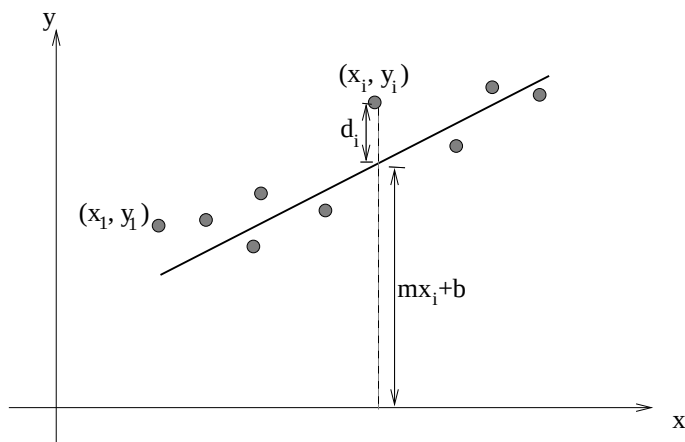
$$\nabla f(x, y) = \lambda \nabla g(x, y), \text{ where } g(x, y) = x^2 + 4y^2 - 1.$$

$$\Rightarrow (-ye^{-xy}, -xe^{-xy}) = \lambda(2x, 8y) \quad (1)$$

If $x = 0$ then $y = 0$, however $(0, 0)$ does not satisfy $x^2 + 4y^2 = 1$ (also $(0, 0)$ is a saddle point as noted earlier).

If $x \neq 0$ then $y \neq 0$ and we get $\frac{y}{2x} = \frac{x}{8y}$ or $x^2 = 4y^2$, which gives $x = \pm \frac{1}{\sqrt{2}}$ and $y = \pm \frac{1}{2\sqrt{2}}$.

Hence the four possible solutions of (1) are $(x, y) = (\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{2\sqrt{2}})$.



Since $f(\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}) = f(-\frac{1}{\sqrt{2}}, -\frac{1}{2\sqrt{2}}) = e^{-\frac{1}{4}} < f(-\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}) = f(\frac{1}{\sqrt{2}}, -\frac{1}{2\sqrt{2}}) = e^{\frac{1}{4}}$.
 $(\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}})$ and $(-\frac{1}{\sqrt{2}}, -\frac{1}{2\sqrt{2}})$ are absolute minima of f and
 $(-\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}})$ and $(\frac{1}{\sqrt{2}}, -\frac{1}{2\sqrt{2}})$ are absolute maxima of f .

8. Suppose that a scientist has reason to believe that two quantities x and y are related linearly, that is, $y = mx + b$, at least approximately for some values of m and b . The scientist performs an experiment and collects data in the form of points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, and then plots these points. The points don't exactly lie on a straight line, so the scientist wants to find constants m and b such that the line $y = mx + b$ "fits" the points as well as possible. Let $d_i = y_i - (mx_i + b)$ be the vertical deviation of the point (x_i, y_i) from the line. The **method of least squares** determines m and b so as to minimize $\sum_{i=1}^n d_i^2$, the sum of the squares of these deviations. Show that according to this method, the line of best fit is obtained when

$$m \sum_{i=1}^n x_i + bn = \sum_{i=1}^n y_i$$

$$m \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i = \sum_{i=1}^n x_i y_i.$$

Solution: To find m and b such that $f(m, b) = \sum_{i=1}^n d_i^2$ is minimum, we have to find

the critical points of f

$$f_m(m, b) = 0 \text{ gives } \sum_{i=1}^n d_i(-x_i) = 0 \quad (1)$$

$$\text{and } f_b(m, b) = 0 \text{ gives } \sum_{i=1}^n d_i = 0 \quad (2)$$

From (2) we get

$$m \sum_{i=1}^n x_i + bn = \sum_{i=1}^n y_i.$$

From (1) we get

$$m \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i = \sum_{i=1}^n x_i y_i.$$

9. Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified line.

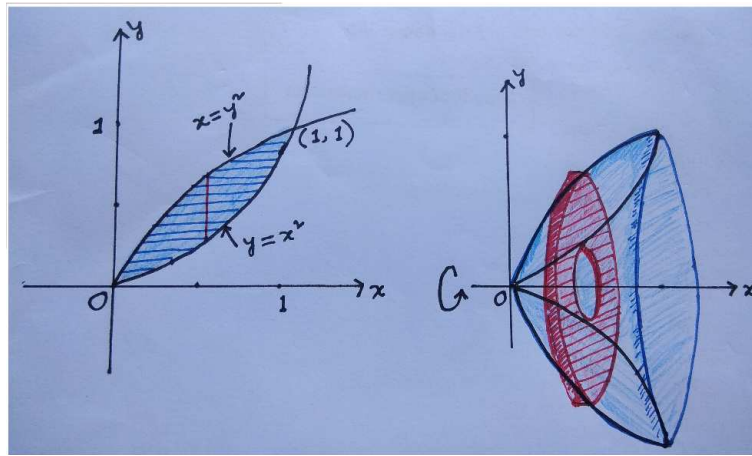
- (a) $y = x^2$, $y^2 = x$; about x -axis
- (b) $y^2 = x$, $x = 2y$; about y -axis
- (c) $y = x$, $y = x^2$; about the line $x = -1$.

Solution:

(a) For a given x the area of a cross section of the solid obtained is given by $A(x) = \pi(\sqrt{x})^2 - \pi(x^2)^2$.

Hence the volume of the solid is given by

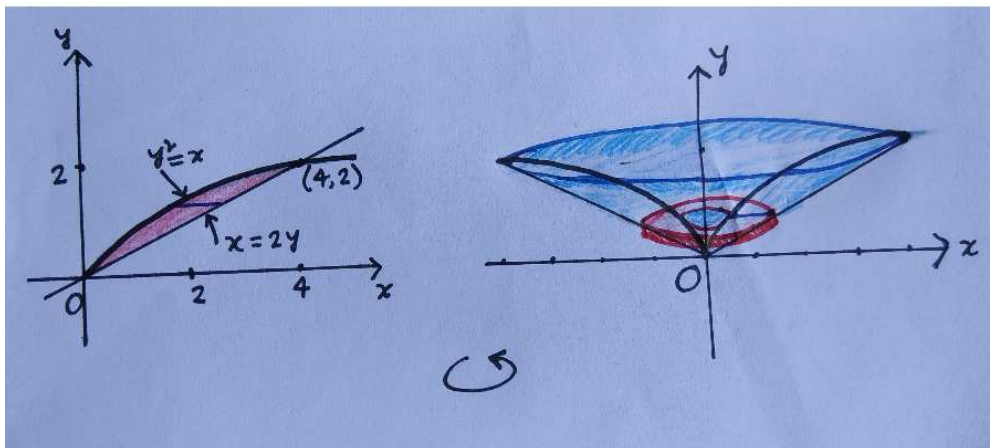
$$V = \int_0^1 A(x)dx = \frac{3\pi}{10}.$$



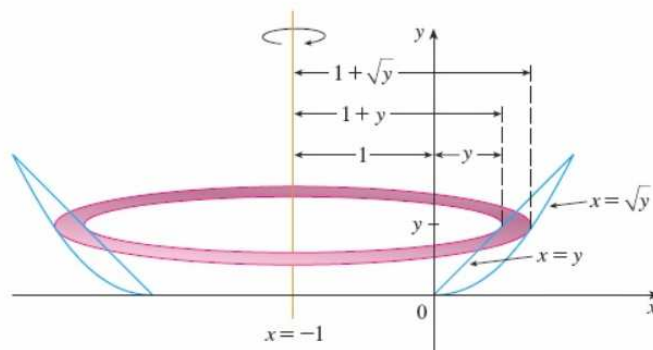
(b) For a given y the area of a cross section of the solid obtained is given by $A(y) = \pi(2y)^2 - \pi(y^2)^2$.

Hence the volume of the solid is given by

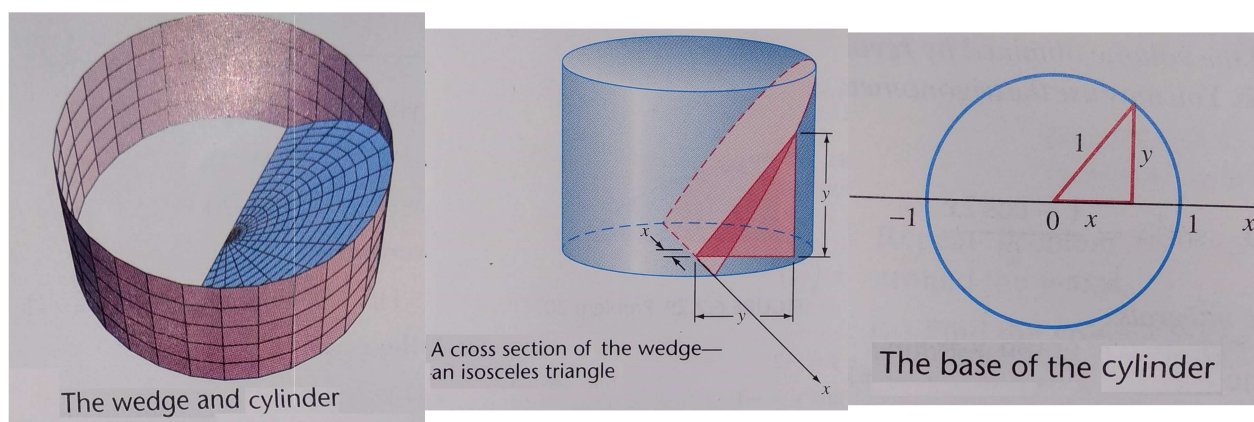
$$V = \int_0^2 A(y)dy = \frac{64\pi}{15}.$$



(c) For a given y the area of a cross section of the solid obtained is given by $A(y) = \pi(\sqrt{y} + 1)^2 - \pi(y + 1)^2$.
Hence the volume of the solid is given by $V = \int_0^1 A(y)dy = \frac{\pi}{2}$.



10. Find the volume of the wedge that is cut from a circular cylinder with unit radius and unit height by a plane that passes through a diameter of the base of the cylinder and through a point on the circumference of its top.



Solution: We show the cylinder and the wedge in the figure above.

To form such a wedge, you may fill a cylindrical glass with water and then drink slowly, tipping the bottom up as you drink, until the half of the bottom of the glass is exposed; the remaining water forms the edge.

Without loss of generality let us assume the plane to cut the base of the cylinder along the x axis, with the centre of the cylinder at the origin (see the second figure in the sequence).

Then for a fixed x the cross section of the wedge is an isosceles triangle, since the plane intersects the base of the cylinder at a fixed angle, and at $x = 0$, the height and the base (given by y) are both equal to 1 (see the third figure in the sequence).

The area of each of these triangles is given by:

$$A(x) = \frac{1}{2}y \times y = \frac{1}{2}(1 - x^2).$$

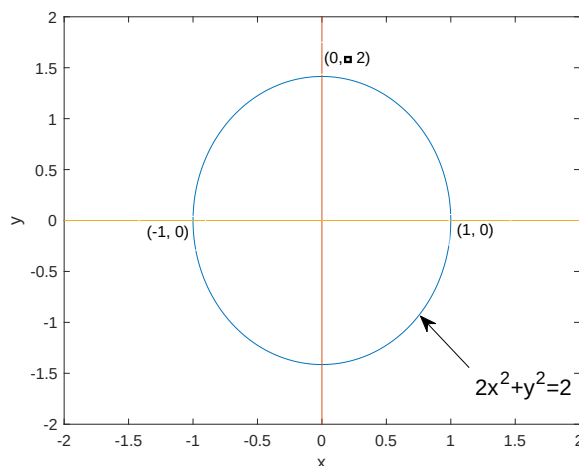
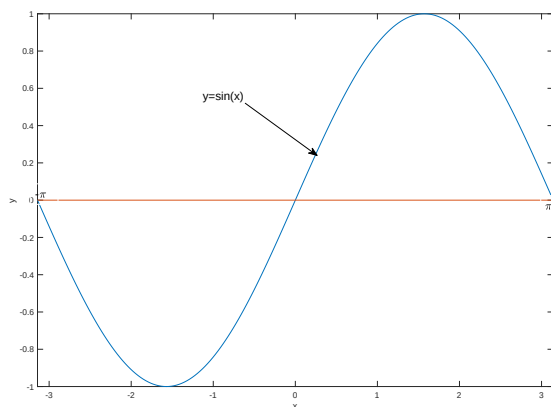
Hence the required volume is

$$V = \int_{-1}^1 A(x)dx = \int_{-1}^1 \frac{1}{2}(1 - x^2)dx = \frac{2}{3}.$$

11. Prove that the length of one arch of the sine curve $y = \sin x$ is equal to half the circumference of the ellipse $2x^2 + y^2 = 2$.

Solution: The length of one arch of the sine curve is given by:

$$\begin{aligned} L_1 &= \int_0^\pi \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\ &= \int_0^\pi \sqrt{1 + (\cos x)^2} dx. \end{aligned}$$



The length of half the circumference of the ellipse is given by:

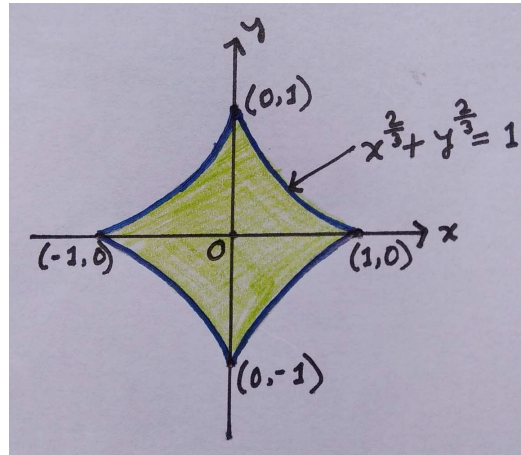
$$\begin{aligned} L_2 &= \int_{-1}^1 \sqrt{1 + \left(\frac{2x}{\sqrt{2-2x^2}}\right)^2} dx \\ &= \int_{-1}^1 \sqrt{1 + \left(\frac{2x}{\sqrt{2-2x^2}}\right)^2} dx \\ &= \int_{-1}^1 \sqrt{\frac{1+x^2}{1-x^2}} dx. \end{aligned}$$

Take $x = \cos t$ then the value of the above integral is equal to:

$$\begin{aligned} &\int_\pi^0 \sqrt{\frac{1 + (\cos t)^2}{(\sin t)^2}} (-\sin t) dt \\ &= \int_0^\pi \sqrt{1 + (\cos t)^2} dt, \end{aligned}$$

hence $L_1 = L_2$.

12. Find the total length of the asteroid given by $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1$ and then find the area of the surface generated by revolving the asteroid around the y -axis.



Solution: By implicit differentiation:

$$\frac{2}{3}x^{-\frac{1}{3}} + \frac{2}{3}y^{-\frac{1}{3}} \frac{dy}{dx} = 0, \text{ provided } x \neq 0 \text{ and } y \neq 0.$$

$$\Rightarrow \frac{dy}{dx} = -\left(\frac{y}{x}\right)^{\frac{1}{3}} \quad \left(\frac{dx}{dy} = -\left(\frac{x}{y}\right)^{\frac{1}{3}}\right).$$

The required length of the asteroid is given by:

$$L = 4 \int_0^1 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \left(= 4 \int_0^1 \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy\right)$$

$$L = 4 \int_0^1 \sqrt{1 + \left(-\left(\frac{y}{x}\right)^{\frac{1}{3}}\right)^2} dx$$

$$= 4 \int_0^1 x^{-\frac{1}{3}} dx = 6.$$

As the asteroid is symmetric about the origin we can assume $y \geq 0$, then the required surface area is given by:

$$S = 2 \int_0^1 2\pi x ds = 4\pi \int_0^1 x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$= 4\pi \int_0^1 x \sqrt{1 + \left(-\left(\frac{y}{x}\right)^{\frac{1}{3}}\right)^2} dx$$

$$= 4\pi \int_0^1 x \times x^{-\frac{1}{3}} dx$$

$$= 4\pi \int_0^1 x^{\frac{2}{3}} dx$$

$$= 4\pi \times \frac{3}{5} \left[x^{\frac{5}{3}}\right]_0^1 = \frac{12\pi}{5}.$$

$$\begin{aligned}
\textbf{Aliter: } S &= 2 \int_0^1 2\pi x ds = 4\pi \int_0^1 (1 - y^{\frac{2}{3}})^{\frac{3}{2}} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy \\
&= 4\pi \int_0^1 (1 - y^{\frac{2}{3}})^{\frac{3}{2}} y^{-\frac{1}{3}} dy \\
&= 4\pi \int_{\frac{\pi}{2}}^0 (\sin \theta)^3 (-3 \cos \theta \sin \theta) d\theta \quad (\text{by taking } y^{\frac{2}{3}} = \cos^2 \theta). \\
&= 12\pi \int_0^{\frac{\pi}{2}} (\sin \theta)^4 (\cos \theta) d\theta = \frac{12\pi}{5}.
\end{aligned}$$